# Bachelorproefvoorstelling van grafenkleuringen

## A computational approach for variants of coloring planar graphs

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# Overzicht


1 Inleiding

2 Literatuur

3 Methodologie

4 Planning

5 Demo

6 Vragen

## Onderzoeksprobleem

- Hoe kunnen we een efficiënt algoritme voor het kleuren van vlakke grafen maken, zodat we hiermee vermoedens kunnen testen.
- Kleuren van vlakke grafen
  - Knopen
  - Vlakke grafen
- Vermoedens rond chromatische getallen weerleggen of ondersteunen door computationeel bewijs

# Grafenkleuringen

- Proper colouring

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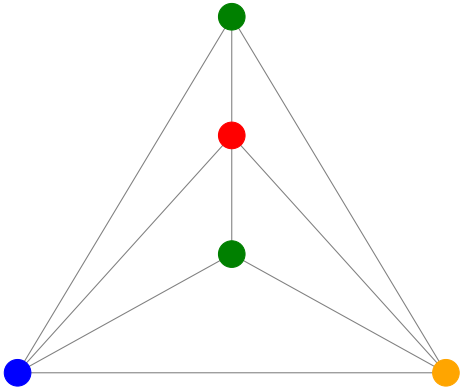

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  - "Oneven"kleur

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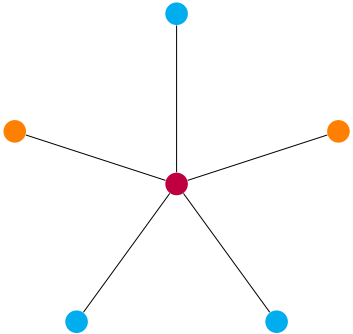

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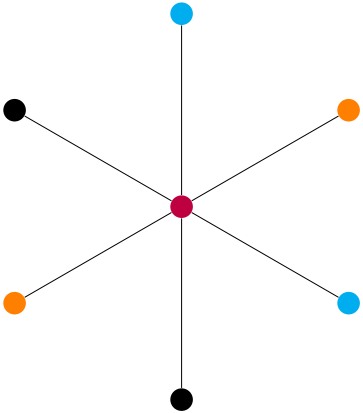

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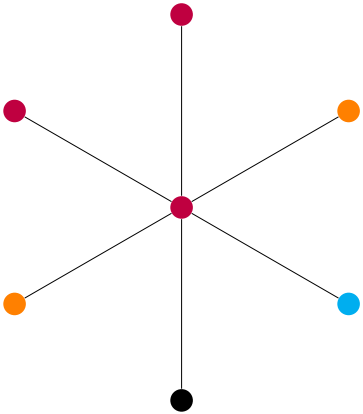

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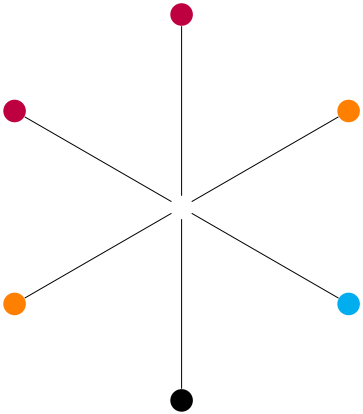

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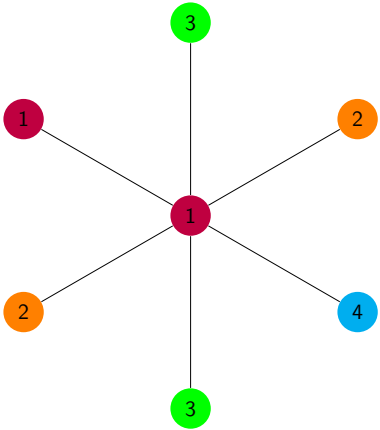

## Toepassingen

- Draadloze netwerken: versturen signaal van zendmast zonder interferentie
  $\rightarrow$ Conflict free colouring
- Vertex ranking: kleuring waarbij paden tussen knopen met zelfde kleur een knoop bevatten met grotere kleur
  $\rightarrow$ Unique-Maximum colouring
  - Toepassingen bij parallel factoriseren matrices en meer

## Naamgeving

- Proper $\rightarrow$ p
- Improper $\rightarrow$ i
- Open neighbourhood $\rightarrow$ o
- Closed neighbourhood $\rightarrow$ c
- pCFo, iUMo, iUMc, pUMo, iCFc, ...
- 10 kleuringen

## Chromatische getallen

- $\chi_{pCFo}(\mathcal{P})$, $\chi_O(\mathcal{P})$, $\chi(\mathcal{P})$, ... ($\mathcal{P}$ een vlakke graaf)
- Per kleuring
- Maximaal aantal kleuren dat minimaal nodig is

## Literatuur

- Brengen resultaten samen en bewijzen nieuwe grenzen[1][2][3]
- Formuleren vermoedens[1][2]

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[1] Igor Fabrici e.a. *Proper conflict-free and unique-maximum colorings of planar graphs with respect to neighborhoods*. Feb 2022. DOI: 10.48550/arXiv.2202.02570

[2] Mirko Petruševski en Riste Škrekovski. "Colorings with neighborhood parity condition". In: *Discrete Applied Mathematics* 321 (2022), p. 385–391. ISSN: 0166-218X. DOI: https://doi.org/10.1016/j.dam.2022.07.018. URL: https://www.sciencedirect.com/science/article/pii/S0166218X22002803

[3] Jan Petr en Julien Portier. *Odd chromatic number of planar graphs is at most 8*. Jan 2022. DOI: 10.48550/arXiv.2201.12381

## Vermoedens

$\mathcal{P}$ een vlakke graaf

### Odd colouring

$$5 \leq \chi_{\mathrm{O}}(\mathcal{P}) \leq 8$$

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$\mathcal{P}$ een vlakke graaf

## Odd colouring

$$5 \leq \chi_{\mathrm{O}}(\mathcal{P}) \leq 8 \qquad \Rightarrow \qquad \chi_{\mathrm{O}}(\mathcal{P}) = 5$$

## Proper colourings

$$6 \leq \chi_{\mathrm{PCFo}}(\mathcal{P}) \leq 8$$

$$\chi_{\mathrm{pCFc}}(\mathcal{P}) = 4$$

$$6 \leq \chi_{\mathrm{pUMo}}(\mathcal{P}) \leq 10$$

$$6 \leq \chi_{\mathrm{pUMc}}(\mathcal{P}) \leq 8$$

# Vermoedens

$\mathcal{P}$ een vlakke graaf

## Odd colouring

$$5 \leq \chi_{\mathrm{O}}(\mathcal{P}) \leq 8 \quad \Rightarrow \quad \chi_{\mathrm{O}}(\mathcal{P}) = 5$$

## Proper colourings

$$\begin{aligned} &6 \leq \chi_{\mathrm{PCFo}}(\mathcal{P}) \leq 8 \\ &\chi_{\mathrm{pCFc}}(\mathcal{P}) = 4 \\ &6 \leq \chi_{\mathrm{pUMo}}(\mathcal{P}) \leq 10 \\ &6 \leq \chi_{\mathrm{pUMc}}(\mathcal{P}) \leq 8 \end{aligned} \quad \Rightarrow \quad \begin{aligned} &\chi_{\mathrm{PCFo}}(\mathcal{P}) = 6 \\ &\chi_{\mathrm{pUMo}}(\mathcal{P}) \leq 7 \\ &\chi_{\mathrm{pUMc}}(\mathcal{P}) = 6 \end{aligned}$$

# Vermoedens 2

$\mathcal{P}$ een vlakke graaf

## Improper colourings

$$4 \leq \chi_{\mathrm{iCFo}}(\mathcal{P}) \leq 5$$

$$3 \leq \chi_{\mathrm{iCFc}}(\mathcal{P}) \leq 4$$

$$5 \leq \chi_{\mathrm{iUMo}}(\mathcal{P}) \leq 10$$

$$4 \leq \chi_{\mathrm{iUMc}}(\mathcal{P}) \leq 6$$

## Vermoedens 2

$\mathcal{P}$ een vlakke graaf

**Improper colourings**

$$\begin{aligned} 4 &\leq \chi_{\mathrm{iCFo}}(\mathcal{P}) \leq 5 \\ 3 &\leq \chi_{\mathrm{iCFc}}(\mathcal{P}) \leq 4 \\ 5 &\leq \chi_{\mathrm{iUMo}}(\mathcal{P}) \leq 10 \\ 4 &\leq \chi_{\mathrm{iUMc}}(\mathcal{P}) \leq 6 \end{aligned} \quad \Rightarrow \quad \begin{aligned} \chi_{\mathrm{iCFo}}(\mathcal{P}) &= 4 \\ \chi_{\mathrm{iCFc}}(\mathcal{P}) &= 3 \\ \chi_{\mathrm{iUMo}}(\mathcal{P}) &= 5 \\ \chi_{\mathrm{iUMc}}(\mathcal{P}) &= 4 \end{aligned}$$

- Conjecture van $\chi_{\mathrm{iCFc}}(\mathcal{P})$ is al weerlegd[4].
- $\chi_{\mathrm{iCFc}}(\mathcal{P}) = 4$

[4] Lukáš Málik. *Conflict-free colouring of planar graphs [online]*. Bakalářská práce. SUPERVISOR: prof. RNDr. Petr Hliněný, Ph.D.. 2023 [cit. 2025-11-26]. URL: https://theses.cz/id/u5w3fx/.

## Methodologie

- Backtracking
- Houd mogelijke kleuren bij

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**Algorithm 2** Kleuren van grafen

```
G ← grafen gegenereerd door plantri
for g ∈ G do
    for maxColor ≤ 10 do
        if colorGraph(g, maxColor, 0) then
    return maxColor
```

**Algorithm 3** `colorGraph`

```
if All vertices colored then return True
vertex ← vertices[index]
for Each available color of vertex do
    color(vertex, color)
    if neighbourhood(vertex) is fully colored ∧ ¬isCorrectlyColored(vertex) then
        return False
    updateNeighbours(vertex, color)
    if colorGraph(g, maxColor, getBestIndex()) then
        return True
    uncolor(vertex, color)
return False
```

**Algorithm 4** `updateNeighbours`

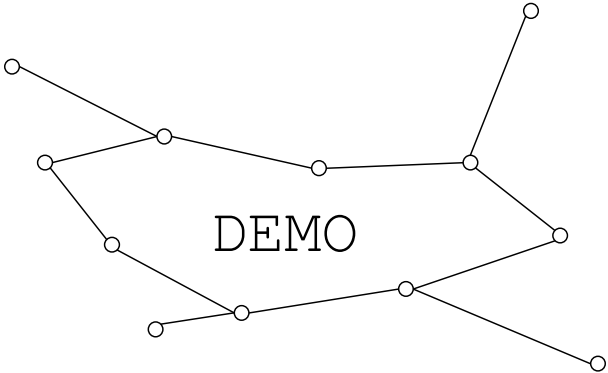
```
for neighbour ∈ neighbourhood(vertex) do
    if neighbourhood(neighbour) is fully colored then
        return isCorrectlyColored(neighbour)
    else if neighbourhood(neighbour) contains only one uncolored vertex then
        removeAvailableColorsForColoring(uncoloredNeighbour)
    else if coloring is proper then
        removeAvailableColor(neighbour, color)
```

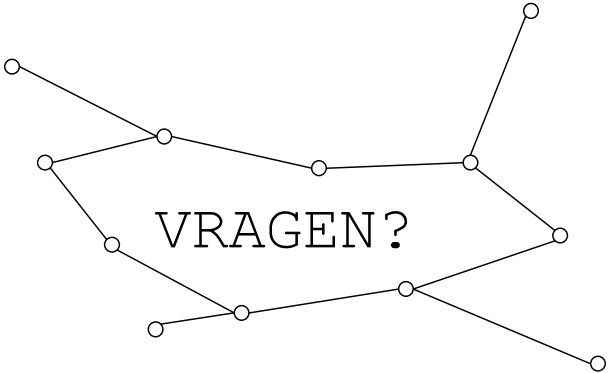
## Threats to validity

- Incorrect algoritme
- Vergelijken tussen implementaties

# Planning

- Optimaliseren
- Patronen in maximaliserende grafen zoeken
- Code omzetten naar C
- Algoritme laten lopen
- Bewijzen correctheid naïef algoritme
- Verslag schrijven


DEMO


VRAGEN?